

Generation Interconnection System Impact Study Report

For Queue Project Q19-1369

Monaco 138 kV

May 30, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1369. The Interconnection Customer proposes a 340.9 MW solar photovoltaic facility interconnecting at the Monaco 138 kV substation in the Pennsylvania area. The request was received on December 27, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 446 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,813,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Alford – Liberty Electric Power Plant 138.0 kV	Liberty Electric Power Plant – Warehouse Electric 138.0 kV L	129.3	192	5.3
Bridgeport – Marcus Hook Energy Lp 138.0 kV Line	Trainer – Bridgeport 138.0 kV	117.9	353	11.2
Liberty Electric Power Plant – Lke Catherine 138.0 kV	Guilbeau Electric 345-138 kV Autotransformer	111.7	514	7.9
Red Rock – Green Country 138.0 kV	Lake Catherine – Bridgeport 138.0 kV	106.9	300	9.0
Reedville – Lafayette Catepillar 138.0 kV Line	Protein – 26Th And F 138.0 kV	104.6	216	7.1
Lake Catherine – Bridgeport 138.0 kV	Guilbeau Electric 345/138 kV Transformer Bank	103.5	555	7.9
Red Rock – Lafayette Catepillar 138.0 kV	Protein – 26Th And F 138.0 kV	100.7	201	7.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Liberty Electric Power Plant – Alfrd 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,957,000	100% (\$1,957,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bridgeport – Marcus Hook Energy Lp 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,159,000	100% (\$1,159,000)
Lake Catherine – Liberty Electric Power Plant 138.0 kV	Rebuild line with higher-capacity conductor	\$4,888,000	78% (\$3,792,000)
Red Rock – Green Country 138.0 kV	Reconductor line	\$526,000	78% (\$412,000)
Reedville – Lafayette Catepillar 138.0 kV	Reconductor line	\$637,000	100% (\$637,000)
Bridgeport – Lake Catherine 138.0 kV Line	Reconductor line	\$525,000	78% (\$407,000)
Lafayette Catepillar – Red Rock 138.0 kV	Reconductor line	\$449,000	100% (\$449,000)
	Total allocated to project		\$8,813,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.